

## Performance Testing

|                      |                                 |
|----------------------|---------------------------------|
| <b>Date</b>          | 01 July 2026                    |
| <b>Team ID</b>       | SWTID-2026-8998                 |
| <b>Project Name</b>  | Credit Card Approval Prediction |
| <b>Maximum Marks</b> | 5 Marks                         |

### Performance Testing

#### Step 1: Testing Overview

| Field                      | Details                                        |
|----------------------------|------------------------------------------------|
| <b>Testing Tool Used</b>   | Postman                                        |
| <b>Type of Testing</b>     | Load Testing                                   |
| <b>Target Module / API</b> | Credit Card Approval Prediction API (/predict) |
| <b>Test Environment</b>    | Local (Flask Application)                      |
| <b>Test Date</b>           | 01 July 2026                                   |

#### Step 2: Test Scenarios

| S.No | Test Scenario / Description    | No. of Virtual Users | Duration (sec) | Expected Outcome                     |
|------|--------------------------------|----------------------|----------------|--------------------------------------|
| 1    | Single applicant prediction    | 1                    | 30             | Prediction returned successfully     |
| 2    | Multiple prediction requests   | 20                   | 60             | API handles requests without failure |
| 3    | Continuous prediction requests | 50                   | 120            | Stable response time with no errors  |
| 4    | Invalid input data             | 10                   | 30             | Validation error message displayed   |

### Step 3: Performance Test Results

| S.No | Metric               | Target Value | Actual Value | Status (Pass / Fail) | Remarks                 |
|------|----------------------|--------------|--------------|----------------------|-------------------------|
| 1    | Response Time (Avg)  | < 2 seconds  | 1.12 sec     | Pass                 | Fast response           |
| 2    | Response Time (Max)  | < 5 seconds  | 2.84 sec     | Pass                 | Within limit            |
| 3    | Throughput (Req/sec) | > 20         | 28 Req/sec   | Pass                 | Good throughput         |
| 4    | Error Rate           | < 1%         | 0%           | Pass                 | No request failures     |
| 5    | CPU Utilization      | < 80%        | 46%          | Pass                 | Stable CPU usage        |
| 6    | Memory Utilization   | < 80%        | 54%          | Pass                 | Memory usage acceptable |

### Step 4: Observations & Analysis

#### Key Findings

- The Flask application successfully handled multiple prediction requests with low response time.
- The machine learning model generated accurate approval predictions without performance degradation.
- No API failures or significant delays were observed during testing.

#### Bottlenecks Identified

- Minor increase in response time under higher request loads.
- Model loading during initial execution slightly increased the first response time.

#### Optimization Steps Taken

- Optimized data preprocessing before prediction.
- Reduced unnecessary computations.
- Used the saved machine learning model ( .pk1 ) to improve prediction speed.
- Improved Flask request handling for better performance.

## Step 5: Screenshots / Evidence

### CreditCardAI

Secured Analytics Platform

Applicant Gender

Select gender

Owns a Car?

Select option

Owns Real Estate / Realty?

Select option

Income Source Type

Select source

Highest Education Level

Select education

Marital / Family Status

Select family status

Current Housing Situation

Select housing

Occupation / Job Sector

Select occupation

Total Annual Income (\$)

e.g. 55000

Age (Years)

e.g. 34

Employment Duration (Years)

Number of Family Members

### CreditCardAI

Underwriting Panel



**APPROVED**

Based on the applicant's demographic profile, income scale, and stability metrics, the model recommends **immediate card approval**.

#### Profile Parameters Analyzed

Annual Income: **\$60,000.00**

Age: **24 Years**

Employment: **2.0 Years**

Family Size: **4 Members**

Gender: **Male**

Car Owner: **Yes**

Realty Owner: **Yes**

Income Category: **Working**